

How I Work

As an Engineer

What's the problem?

- /s it a problem?
- What was the expectation?
- What is the desired outcome?
- Does proposed solution solve the perceived problem?

A '*problem*' can be many things,
'*intention*' is easier to define.

Build a 'Student Registry'

Brief:

- SPA: a list with full name & DOB in specific format
- When a student is selected, another view (of your choice) will display additional details
- ONLY First & Last name should be editable, rest of the details are read only

Provided:

- Data set
- API endpoints

Design Considerations

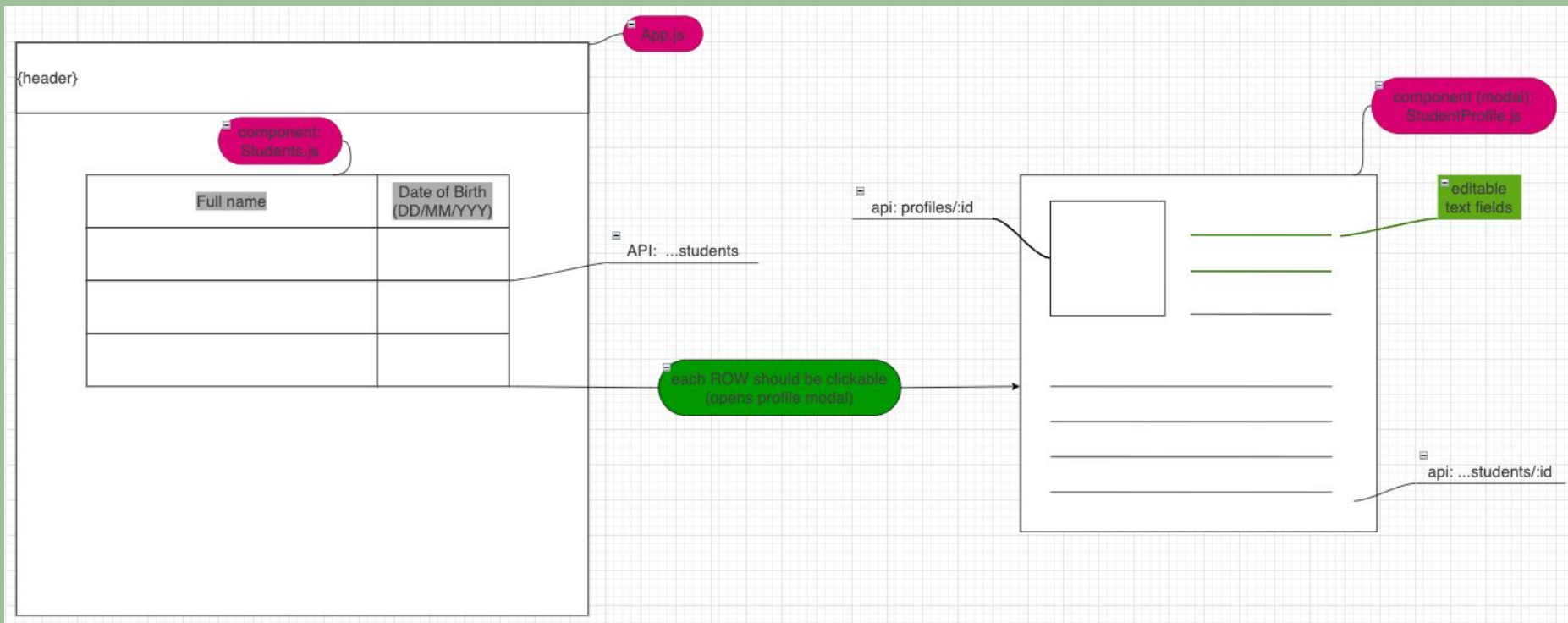
Core Requirements:

- List
- 'Another view' for details
 - Decided on Modal instead of new page because:
 - Requirements only intended for name to be editable
 - Small amount of data as details
- 2x editable fields

Approach: Wireframe

- Helps me visualise **& plan** what I'm building
- Prevents scope creep & keeps me focused on core requirements

➔ wireframe.drawio.svg



Key Technical Features

- CRA vs Vite
 - Presumed little difference for small project
 - But 8gb ram struggled with hot reloading etc
- Material UI
 - Familiarity
 - Good documentation
 - More than sufficient for my needs
- Jest
 - Was not a requirement
 - Personally felt it was (is) important
 - Familiarity

Brief overview of process

Started from ground up:

1. App.js
2. Student Register
3. date formatter
4. Student Profile
5. Tests
6. Hooked StudentProfile up with StudentRegister

Challenges, Problem-solving & Learnings

- No copilot back then!
- Stack Overflow, medium
- Documentation
- Console.log
- Wireframe

Future Improvements

- Implementing TypeScript
- Fully implement loading and error states
- More tests
 - Unit test eg. date formatter
 - Integration test eg. StudentProfile only rendered when a row in the StudentRegister is clicked.
 - E2E eg. user can complete the flow from opening a student profile and updating the name.